

LSAPy: Land Suitability Analysis in Python

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Summary

LSAPy is a highly customizable Python library designed to streamline and enhance Land Suitability Analysis (LSA) workflows. The package implements a fuzzy-logic approach and provides two core objects - `SuitabilityCriteria` and `LandSuitabilityAnalysis` - and one module (`lsapy.standardize`) that work together to deliver a flexible and user-defined LSA framework. By relying on xarray objects for computation (Hoyer & Hamman, 2017), LSAPy seamlessly integrates with the broader Python ecosystem, such as dask for efficient parallel processing and matplotlib for data visualisation. Its modular design addresses some limitations of existing LSA tools by offering greater flexibility, reproducibility, and scalability for research and practical applications.

Statement of need

In the past decades, several software programs have been developed to perform land evaluation or suitability analysis, including ALES (Johnson & Cramb, 1991), Micro-LEIS (D. De La Rosa et al., 1992; D. De La Rosa et al., 2004), LEIGIS (Kalogirou, 2002), ALSE (Elsheikh et al., 2013), and general-purpose GIS platforms (e.g., ArcGIS and QGIS) also provide functionalities for this purpose. While each of these tools offers distinct advantages, a limitation that often arises is the software-imposed constraints and the lack of freedom left to users (Asaad et al., 2022; Chen et al., 2022; Elsheikh et al., 2013). These limitations manifest in several ways. Desktop GIS solutions, for example, often present challenges related to operating system dependency, the cost of proprietary software (e.g., ArcGIS), and difficulties in integration with broader analytical frameworks (Chen et al., 2022). Furthermore, many specialised programs (e.g., Micro-LEIS, LEIGIS, ALSE) do not allow modification of the land characteristics used in the analysis (Asaad et al., 2022; Elsheikh et al., 2013). This rigidity is problematic, as the predefined land characteristics may not apply to all crops or may require other parameters. Additionally, some tools (e.g., LEIGIS) support only a limited selection of crops for evaluation (Elsheikh et al., 2013).

Recently, two libraries – ALUES (Asaad et al., 2022) and PyLUSAT (Chen et al., 2022) – have been developed to address some of these limitations. ALUES, an R package, supports the evaluation of 56 crops and allows the addition of new ones. While it addresses some of the limitations discussed above, it falls short in others. For instance, it relies on fixed criteria grouped into three categories (i.e., land and soil, water, and temperature) that are unmodifiable (Asaad et al., 2022). PyLUSAT, a Python package, enables land suitability analysis using vector data, offering specific advantages such as bypassing the modifiable areal unit problem (MAUP) and the political relevance of object shape (Chen et al., 2022). Nevertheless, raster data models have been recognized as more appropriate for LSA applications because of their area-oriented structure (Malczewski, 2004). Moreover, raster-based approaches are generally more efficient for data combinations and complex calculations (Carr & Zwick, 2007), particularly when integrating climate data, which is often distributed in netCDF format and best processed using

43 raster routines. Converting such data to a vector format for analysis is technically feasible but
44 suboptimal, leading to increased computational overhead and memory usage. Consequently,
45 raster-based methods are essential for large-scale analyses, such as land suitability assessments
46 using climate projections.

47 Beyond these technical constraints, the lack of user flexibility in existing software introduces
48 further limitations. First, when programs offer a restricted set of functionalities, they hinder the
49 reproducibility of scientific results. For instance, ALSE aggregates criteria suitability using the
50 maximum limitation method (Elsheikh et al., 2013), whereas ALUES provides only minimum,
51 maximum, and average aggregation options (Asaad et al., 2022). This discrepancy renders the
52 reproduction of cross-software results impossible. Second, tools designed for specific use cases
53 (e.g., agriculture) limit broader adoption and, in the case of free and open-source software
54 (FOSS), may impede community engagement and growth. Finally, rigid frameworks restrict
55 the ability to explore edge cases or out-of-the-box analyses, which are increasingly important
56 in a rapidly changing environment.

57 The limitations of existing software programs motivated the development of LSAPy (Land
58 Suitability Analysis in Python), a new, highly customizable Python package supporting array-like
59 data, such as raster and netCDF formats. Thanks to its customisation approach, frameworks
60 used in most software programs mentioned above can be implemented in LSAPy.

61 Land Suitability Analysis Workflow

62 LSAPy provides two core objects and one module that operate together to perform the land
63 suitability analysis (LSA) according to user-defined frameworks (Figure 1).

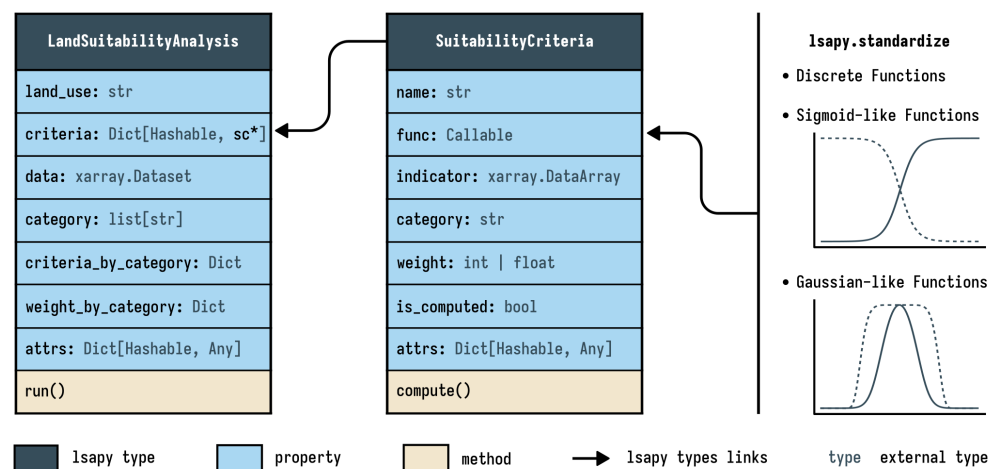


Figure 1: Overview of LSAPy's objects structures and their associated properties and methods. *sc is used as an abbreviation of SuitabilityCriteria.

64 Standardize Module

65 lsapy.standardize provides a set of standardization functions used to transforms input data
66 (e.g., slope, annual mean temperature, annual total precipitation...) into standardized scale
67 of suitability values ranging from 0 to 1 (Figure 1). LSAPy includes built-in functions for
68 both discrete and continuous data. For the latter, the package follows a fuzzy-logic approach,
69 implementing sigmoid-like and Gaussian-like membership functions previously used in LSA
70 studies.

Suitability Criteria

The SuitabilityCriteria defines an individual criteria used in LSA. Its func property refers to the standardization function, while indicator specifies the input data (Figure 1). The weight and category properties allow users to customise how each criteria is aggregated with others in the analysis. is_computed indicates whether the indicator already corresponds to suitability values and does not need further computation. Finally, the compute() method applies the standardization function to the given indicator to calculate the suitability score.

Land Suitability Analysis

LandSuitabilityAnalysis is the top-level class in LSAPy, defining the LSA framework (Figure 1). All criteria for the analysis are stored in the criteria property. The category, criteria_by_category and weights_by_category properties are populated based on the defined criteria with the first one listing all unique categories, and the latter two their associated criteria and weights respectively. The run() method executes the LSA, with parameters specifying the level of suitability to compute (i.e., criteria, category, or overall land suitability) and the aggregation method to use. Currently, supported aggregation methods include median, mean, weighted mean, geometric mean, weighted geometric mean, and limiting factor.

Additional Features

- The stats module offers functions for computing spatial (e.g., national, regional...) summary statistics of land use suitability.
- LSAPy's open_data function provides access to sample datasets, including soil/land data from NZGLID (Hamon, 2025) and climate data from NEX-GDDP-CMIP6 (Thrasher et al., 2022) for tutorial and training purposes.

Example Workflow

For this example, a land suitability analysis was performed using the soil and climate data provided in LSAPy, covering the New Zealand territory. Three suitability criteria were defined: slope, annual precipitation excess, and the growing temperature requirement. The first criteria was categorised under "soilTerrain", while the latter two were grouped under "climate". The indicators associated with each criteria were the slope (in degrees), annual total precipitation (in mm), and annual mean temperature (in °C) respectively, and are presented in (Figure 2). It is important to note that the chosen criteria, indicators and standardisation functions are for demonstration purposes only and do not correspond to an actual land suitability analysis for any specific crop. A comprehensive LSA would typically involve a larger number and more complex criteria.

The suitability of each criteria was computed by applying the defined standardisation functions ((Figure 2)). The criteria suitability were then aggregated: first, the suitability of the "soilTerrain" and "climate" categories were calculated, and then the overall land suitability was derived by combining the two categories. In this simple example, the arithmetic mean was used for all aggregation steps. The resulting land suitability map is also presented in (Figure 2) and allows identifying the most suitable areas based on the defined criteria.

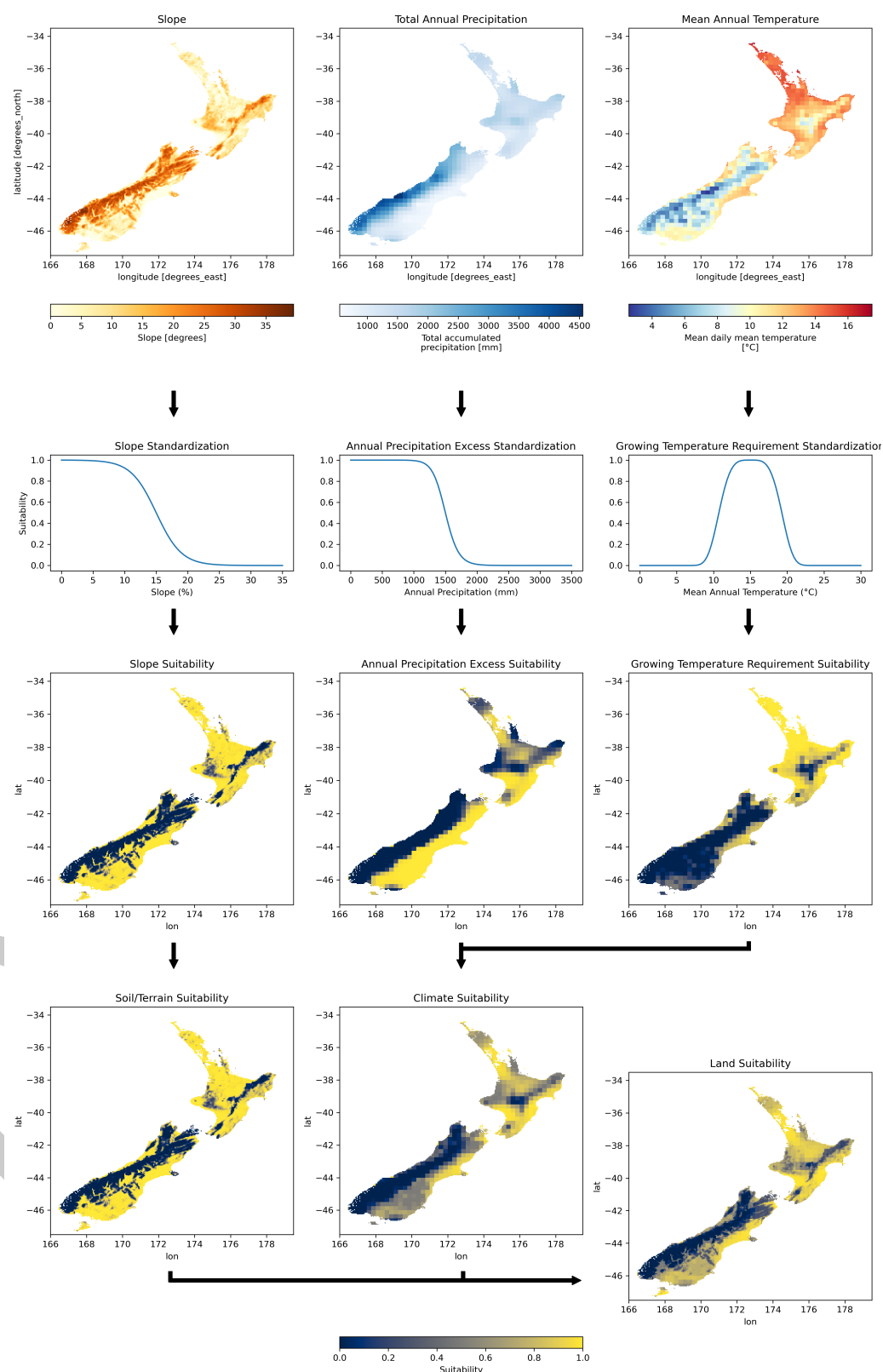


Figure 2: Workflow of the land suitability analysis example. Presented are the input indicators associated with the criteria used, the standardisation functions applied and the output suitability. The suitability of each of the criteria is shown, as well as the aggregated suitability of the categories and the final land suitability.

Research Applications

LSAPy has been used to assess the impact of climate change on apples, cherries, maize and wheat in New Zealand (Hamon et al., 2025a; Hamon et al., 2025b), and is the core component of the New Zealand Land Use Suitability Database.

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